EVERY FORMAL OBJECT OF THE CURVES STACK IS EFFECTIVE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Section artin-section-formal-objects (canonical tag 07X3). cited at proof line 4976. Definitions of formal objects and effectivity.
- Lemma spaces-more-morphisms-lemma-formal-algebraic-space-proper-reldim-1 (canonical tag 0E7R). cited at proof line 4981. Existence of the projective X with compatible identifications.
- Lemma more-morphisms-lemma-check-flatness-on-infinitesimal-nbhds (canonical tag 0D4G). cited at proof line 4986. Flatness via infinitesimal neighbourhoods.
- Lemma more-morphisms-lemma-dimension-fibres-proper-flat (canonical tag 0D4J). cited at proof line 4988. Relative dimension <= 1.

Source reference. Original section 07X3: sources/stacks-chow/tags.txt, source lines 10063–10063. Cited at proof line 4976: definitions.

Source reference. Original lemma 0E7R: sources/stacks-chow/tags.txt, source lines 17791–17791. Cited at proof line 4981: existence of X.

Source reference. Original lemma 0D4G: sources/stacks-chow/tags.txt, source lines 16451–16451. Cited at proof line 4986: flatness.

Source reference. Original lemma 0D4J: sources/stacks-chow/tags.txt, source lines 16454–16454. Cited at proof line 4988: relative dimension <= 1.

Lemma 0.2. *Consider the stack $\mathcal{C}urves$ over the base scheme* $\mathrm{Spec}(\mathbf{Z})$. *Then every formal object is effective.*

Proof. For definitions of the notions in the lemma, please see Artin's Axioms, Section 07X3. Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian complete local ring. Let $(X_n \to \mathrm{Spec}(A/\mathfrak{m}^n))$ be a formal object of $\mathcal{C}urves$ over A. By More on Morphisms of Spaces, Lemma 0E7R there exists a projective morphism $X \to \mathrm{Spec}(A)$ and a compatible system of ismomorphisms $X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A/\mathfrak{m}^n) \cong X_n$. By More on

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Morphisms, Lemma 0D4G we see that $X \to \mathrm{Spec}(A)$ is flat. By More on Morphisms, Lemma 0D4J we see that $X \to \mathrm{Spec}(A)$ has relative dimension ≤ 1. This proves the lemma. $\square$